

OUTPUT FREQUENCIES

$R = R1 + VR1 = R2 + VR2$	$C = C1 = 3.3nF$ (3300pF)	$C = C2 = 33nF$ (33000pF)
10k	10kHz	1kHz
2.2k	45.45kHz	4.54kHz

PARTS LIST

Semiconductors:

IC1 - ICL8038 function generator

Resistors (all 1/4-watt, $\pm 5\%$ carbon):R1, R2 - 2.2k Ω R3 - 10k Ω R4 - 82k Ω VR1, VR2 - 10k Ω potentiometer

Capacitors:

C1 - 3.3nF (3300pF) ceramic disk

C2 - 33nF (33000pF) ceramic disk

Miscellaneous:

CON1-CON4 - 2-pin connectors

S1 - Single-pole, double throw (SPDT) switch

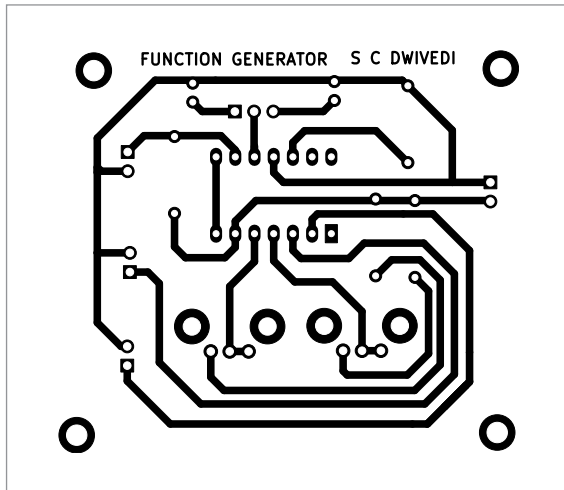


Fig. 3: Actual size PCB

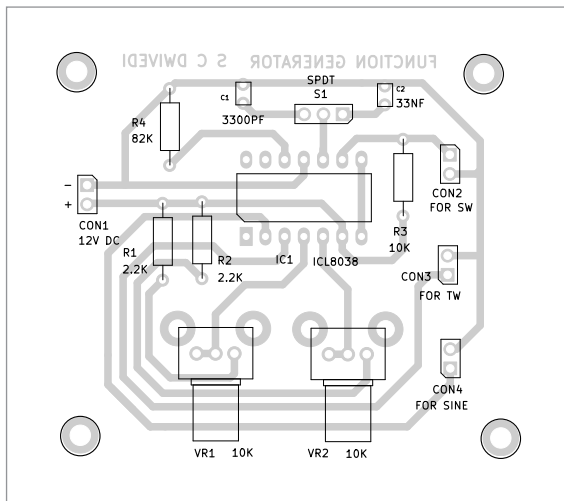


Fig. 4: Components layout

converts the triangular wave into a sine wave by providing a decreasing shunt impedance as the voltage moves towards its minimum and maximum values.

The output frequency is adjustable by changing the resistor values between pin 4 and pin 5 of the IC and the supply. These resistor

values determine the rise and fall times of the sine and triangular waves, as well as the high state of the square wave. Fixed resistors in series with variable resistors ensure that the charging current remains within an optimal range. The resistors' values should not be less than 2.2k Ω for stable operation.

If $R = (R1 + VR1) = (R2 + VR2)$, and $C = C1$ or $C2$ based on switch S1 positions, the output frequencies are as shown in the table above. A wide range of RC combinations can achieve the desired frequency.

The frequency and duty cycle can be adjusted by varying the values of R and C using potentiometers VR1, VR2, and switch S1. For a 50% duty cycle, $R1 + VR1$ should equal $R2 + VR2$. The time

and frequency are independent of the supply voltage. Resistor R3 is a pull-up resistor for generating square waves, while resistor R4 minimises sine wave distortion.

Construction and testing

An actual-size, single-sided PCB layout for the circuit is shown in

Fig. 3, with its component layout illustrated in Fig. 4. After assembling the circuit on the PCB, enclose it in a suitable box.

The device can also be constructed on a general-purpose board. The IC can be powered by a single 12V or dual $\pm 6V$ supply. If a single supply is used, the output will range between the ground and the supply voltage. With a dual supply, all waveforms are symmetrical above and below the ground.

The pull-up resistor R3 should be connected to the power supply if a square wave with a voltage equal to the supply voltage is required. If a square wave of a different voltage is needed, R3 can be connected to an alternate supply, such as +5V, provided that the voltage does not exceed the IC's maximum supply voltage of 30V or $\pm 15V$.

CON1 connects the 12V input power supply to the circuit. Square, triangular, and sine wave outputs are available at CON2, CON3, and CON4, respectively. **EFY**

Pradeep Vasudeva is a member of the Indian Forest Service, presently posted as Director, State Forest Research Institute, Jabalpur. Electronics has been his passion since adolescence, and his areas of interest include amateur radio, RF circuits, and audio projects.

